Efficient Design of Nearfield Focused Beamformer with Frequency Invariant Patterns for Vertical Line Array

Peng Hou, Chunyan Xia and Wenshuai Liu
Dalian Scientific Test & Control Technology Institute, Dalian 116013, China
andyhope@126.com

***Abstract*—This paper proposes an efficient design method of nearfield focused beamformer with frequency invariant patterns for underwater radiated noise measurement, where the impinging of the wideband radiated source on the acoustic array is restricted by the angular and range conditions of the Fresnel approximation. With the Fresnel approximation, the received signal can be first amplitude-compensated and phase-shifted according to the signal attenuation and the wave curvature before the beamforming operation, and then the problem of nearfield wideband beamforming can be readily transformed to a corresponding farfield issue, and the well-established farfield design method of frequency invariant beamformers (FIBs) can be used. Representative example is presented to demonstrate the reduced design complexity and the encouraging beamforming performance under nearfield condition.**

***Index Terms*—Nearfield focused beamforming, vertical line array, frequency invariant beamformer (FIB), Fresnel approximation, radiated-noise measurement.**

I. Introduction

In the field of underwater radiated noise measurement, the vertical line array (VLA) is widely used for enhanced performance in signal detection and interference suppression [1-3]. Here, an efficient design of a wideband beamformer with frequency invariant patterns is one of the most important issues due to its capability of retaining source signature in the mainlobe region and suppressing background noise and interference from the sidelobe region. Furthermore, the radiated-noise source is usually well within the near field of the VLA, which means that the simplified farfield assumption is not valid and it can result in severe degradation of the array performance [4]. Therefore, the curvature of the signal propagation has to be taken into account which requires the nearfield focused beamforming techniques.

Up until now the methods of nearfield wideband beamforming have been relatively limited [4-7] and some of which did not consider the frequency-invariant characteristics on the whole signal band. In [6], Guo et al. present a nearfield compensation method that applied fractional delay filters to compensate for varied propagation delays due to spherical propagation. Although the proposed method is relatively straightforward to implement, a larger number of array elements are required to ensure the precise desired response, and the results achieved are not very satisfying for arrays with less than 20 elements. In [7], Kennedy et al. present a radial beampattern transformation method which uses spherical harmonics to transform the entire desired nearfield wideband beampattern to a physically equivalent farfield beampattern. However, the solution to the wave equation is complex and how to choose a desired beampattern is not clear.

In this paper, an efficient design of nearfield focused beamformer with frequency invariant patterns is presented, where the Fresnel approximation [8] is adopted to calculate the nearfield propagation delay. Then, the problem of nearfield wideband beamforming can be readily transformed to a corresponding farfield issue, and the well-established farfield design method of frequency invariant beamformers (FIBs) can all be used here. The rest of this paper is organized as follows. Section II briefly addresses the received signal model under nearfield condition and the efficient delay calculation with Fresnel approximation. Section III presents an optimal design procedure of the farfield FIBs. Simulation results for performance testing are demonstrated in Section IV.

II. Nearfield Signal Model and the Fresnel Approximation

A. Received Signal Model for VLA

Consider a VLA of N sensors shown in Fig. 1, with d_n the distance between the nth sensor and the reference point z_r, arbitrarily chosen here as the centre of the acoustic array. We assume that a wideband radiated source $S(f_k)$ is located at a focal distance r from the reference point z_r and with an incident angle θ (relative to the broadside) in the near field of the array. Here, all expressions are derived in frequency domain and the signal measured at the output of the nth sensor can be written as

$$X_n(f_k) = g_n(f_k,\theta,r)S_r(f_k) + V_n(f_k), \quad k = 1, 2, \ldots, K, \quad (1)$$

where the argument f_k denotes the dependence of the array data on different frequency bins, $S_r(f_k)$ is the radiated source component received at the reference point, $V_n(f_k)$ is the additive noise component, and

$$g_n(f_k,\theta,r) = a_n(\theta,r)e^{-\mathrm{j}2\pi f_k \tau_n^{\mathrm{ex}}(\theta,r)}, \quad (2)$$

with $\mathrm{j} = \sqrt{-1}$, and the attenuation $a_n(\theta,r)$ and the exact delay $\tau_n^{\mathrm{ex}}(\theta,r)$ accounting for the spherical wavefronts equal to [4]

$$a_n(\theta,r) = r / r_n(\theta,r), \quad \tau_n^{\mathrm{ex}}(\theta,r) = [r_n(\theta,r) - r]/c, \quad (3)$$

where $r_n(\theta,r) = \sqrt{r^2 + d_n^2 - 2d_n r\sin\theta}$ is the distance from the source to the nth sensor, d_n is the location of the nth sensor,

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and c is the speed of sound propagation in water.

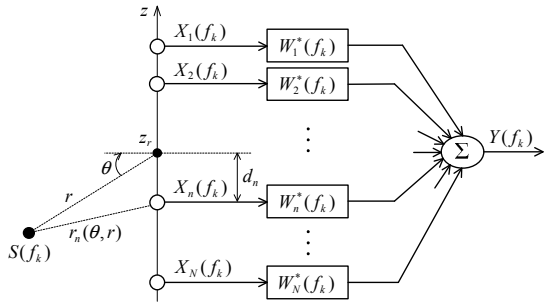


Fig. 1. Configuration of an N-element vertical line array.

The beam response of the acoustic array to a nearfield radiated source can be described as

$$\begin{aligned} b_N(f_k,\theta,r) &= \sum_{n=0}^{N-1} W_n^*(f_k) g_n(f_k,\theta,r) \\ &= \sum_{n=0}^{N-1} W_n^*(f_k) a_n(\theta,r) \exp[-\mathrm{j}2\pi f_k \tau_n^{\mathrm{ex}}(\theta,r)], \end{aligned} \tag{4}$$

where $W_n^*(f_k)$ is the complex weight on the nth sensor, f_k is the operation frequency, and $(\cdot)^*$ denotes the complex conjugate.

Note that in (4), the process of nearfield wideband beamforming requires that different array weights be designed for varied focal distance r, which will greatly increase the computational complexity and post-processing burdens. This is very different from the corresponding process implemented in a farfield circumstance, where the beam response of the array can be written as

$$\begin{aligned} b_F(f_k,\theta) &= \sum_{n=0}^{N-1} W_n^*(f_k) g_n(f_k,\theta) \\ &= \sum_{n=0}^{N-1} W_n^*(f_k) \exp(-\mathrm{j}2\pi f_k d_n \sin\theta / c). \end{aligned} \tag{5}$$

B. Fresnel Delay Approximation

Thanks to the Fresnel expansion [8], an approximation to the exact delay $\tau_n^{\mathrm{ex}}(\theta,r)$ by a formulation that allows an easy on-line computation and implementation can be achieved:

$$\tau_n^{\mathrm{Fr}}(\theta,r) = d_n \sin\theta / c - d_n^2 / 2rc = d_n \sin\theta / c + \tau_n^{\mathrm{cur}}(r). \tag{6}$$

The first term of the addition represents the distance independent delay equal to that defined in (5). The second term also called curvature term takes into account the wave curvature, depends only on the focal distance r.

Making use of (6) we can reformulate (4) as

$$\begin{aligned} b_N(f_k,\theta,r) &= \sum_{n=0}^{N-1} W_n^*(f_k) \exp[-\mathrm{j}2\pi f_k \tau_n^{\mathrm{Fr}}(\theta,r)], \\ &= \sum_{n=0}^{N-1} \Psi_n W_n^*(f_k) \exp(-\mathrm{j}2\pi f_k d_n \sin\theta / c), \end{aligned} \tag{7}$$

where

$$\Psi_n = a_n(\theta,r) \exp[-\mathrm{j}2\pi f_k \tau_n^{\mathrm{cur}}(r)] \tag{8}$$

contains the amplitude-attenuated and phase-shifted terms associated with the nth sensor. It is clear that the received signals can be first compensated with $\Psi_n^C = a_n^{-1}(\theta,r)\Psi_n^*$ according to the signal attenuation and wave curvature before the beamforming operation [6]. After this preliminary correction, the nearfield response can be readily transformed to the farfield, and then the array weights can be designed using the well-established farfield techniques.

Therefore the output signal $Y(f_k)$ can be given by

$$Y(f_k) = \sum_{n=1}^{N} W_n^*(f_k) \Psi_n^C X_n(f_k), \tag{9}$$

where $W_n^*(f_k)$ is the corresponding complex weight obtained under the farfield hypothesis.

However, it should be noted that the restrictive conditions of angular region and range domain must be satisfied to ensure the validity of the Fresnel approximation, which can be written as follows [8]:

$$-18^\circ < \theta < 18^\circ, \quad 1.356 d_{\max} < r < \pi d_{\max}^2 / \lambda, \tag{10}$$

where $d_{\max}$ is the maximum value of $|d_n|$ and λ is the wavelength. In fact, these restrictive conditions (in particular, the angular constraint) are often required in the underwater noise measurement radiated by the naval submarines, where the incident angle θ is restricted by $\theta \le 15^\circ$ [3].

Based on the aforementioned two sections, the design of nearfield focused beamformer has been transformed into a design processing the same structure as in the farfield situation, and a frequency domain design method of FIBs with optimized weight vector $\mathbf{w}(f_k)=[W_1(f_k), W_2(f_k), .., W_N(f_k)]^{\mathrm{T}}$ will be considered and the design details will be discussed in the sequel.

III. Design of Farfield Frequency Invariant Beamformer

Several methods have been proposed to design the FIBs with the simplified farfield model which assumes that all impinging signals at each sensor are plane waves [9-12]. Here, the beamformer design in optimality criterion of least-squares is used, considering the design accuracy and imple-mentation complexity.

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Without loss of generality, we assume that the sensor data have finite frequency band $[f_L, f_H]$, where f_L and f_H are the lower and upper boundary frequencies, respectively, and the frequencies outside this band have been removed before array processing. We discretize the frequency band using a finite number of frequency grids $f_m \in [f_L, f_H]$, $m=1,2,\cdots,M$.

Let f_0 be the reference frequency, which need not be one of $f_m\,(m=1,2,\cdots,M)$. Let $\theta_p \in \Theta_{ML}\,(p=1,2,\cdots,P)$ and $\theta_q \in \Theta_{SL}\,(q=1,2,\cdots,Q)$ be a chosen grid that approximates the mainlobe region Θ_{ML} and the sidelobe region Θ_{SL}, respectively, using finite number of angles.

For a frequency domain processing structure, a typical implementation of FIBs can be stated as [12]

$$\begin{aligned} &\min_{\mathbf{w}(f_m)} \left(\sum_{p=1}^{P} |\, \mathbf{w}^{\mathrm{H}}(f_m)\mathbf{g}(f_m,\theta_p) - b_d(f_0,\theta_p)\,|^2 \right),\ \theta_p \in \Theta_{ML}, \\ &\text{s.t. } |\, \mathbf{w}^{\mathrm{H}}(f_m)\mathbf{g}(f_m,\theta_q)\,| \le \delta,\ \forall \theta_q \in \Theta_{SL}, \\ &\qquad \|\, \mathbf{w}(f_m)\,\| \le \xi, \end{aligned} \tag{11}$$

where

$$\mathbf{g}(f_m,\theta_i) = [g_1(f_m,\theta_i),\, g_2(f_m,\theta_i),\ldots,g_N(f_m,\theta_i)]^{\mathrm{T}}, \tag{12}$$

is the $N\times 1$ farfield manifold vector at frequency bin f_m with $g_n(f_m,\theta_i) = \exp(-\mathrm{j}2\pi f_m d_n \sin\theta_i / c)$, $\theta_i \in \Theta_{ML} \cup \Theta_{SL}$. The desired beampattern $b_d(f_0,\theta_p)$ usually exhibits high gain within the desired spatial sector and yet uniformly low sidelobes in order to suppress unwanted out-of-sector interferences [12]. The parameter δ is a prescribed value to control the sidelobes, and the norm constraint $\|\, \mathbf{w}(f_m)\,\| \le \xi$ limits the white-noise gain to improve the beamformer robustness against random errors in array characteristics [9]. Here $\|\cdot\|$ denotes the Euclidean norm. The optimization problem of (11) can be formulated as a second-order cone programming problem (SOCP) and solved efficiently via the well-established interior point methods, e.g., by SeDuMi MATLAB toolbox [13]. The readers are referred to [14] for detailed solving process.

Therefore, an effeicient design of the nearfield focused beamformer proposed in this paper can be accomplished by the following sequence of steps:

1) Calculate the signal attenuation $a_n(\theta,r)$ (3) and the curvature delay $\tau_n^{\mathrm{cur}}(r)$ (6) according to the focal distance r and incident angle θ. Determine the compensation term Ψ_n^C at operation frequency f_k.

2) Implement the amplitude and phase compensation in frequency domain (9) before the beamforming operation.

3) Design M FIBs (11) that covers the spatial region of interest using farfield beamforming techniques and then weight the received array data in frequency domain (9).

For a varied focal distance r, we only need to recalculate the compensation term Ψ_n^C, and the FIBs under different incident angle θ can be completely calculated offline. Thus, the design complexity and feasibility of the proposed method have been greatly improved.

IV. Restults and Discussion

In this section, we present a representative design example to demonstrate the performance of the proposed method. Consider a vertical acoustic array of $N = 12$ uni-formly spaced sensors, with a $\lambda/2$ spacing at the frequency 1000 Hz, and with a sound speed c = 1500 m/s. Assume that a wideband LFM radiated source $s(t)$ impinges on the array from azimuth $(\theta, r) = (10^{\circ}, 35\mathrm{m})$, which satisfies the restric-tive conditions of the Fresnel approximation. The quantified LFM source signal is given by

$$s(i) = s(t)\,|_{t=i/f_s} = \begin{cases} \sin[2\pi(f_l + \dfrac{f_h - f_l}{2T}t)t], & 0 \le t < T, \\ 0, \quad \text{otherwise.} \end{cases} \tag{13}$$

where f_s = 6250 Hz is the sampling frequency, T is the time interval and the number of snapshots is chosen as $T{\cdot}f_s$ = 512. The parameters f_l and f_h are the lower and upper boundary frequencies chosen as f_l = 500 Hz and f_h = 1000 Hz, respectively. Here, we omit the influence of noise. The waveforms of source signal and its power spectrum are shown in Fig. 2.

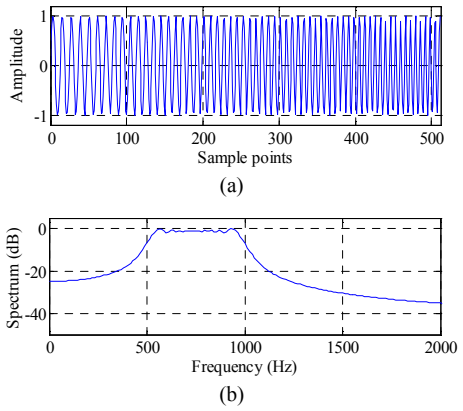


Fig. 2. The source signal (a) and its power spectrum (b).

We then begin our simulation with the given 12-element array to test the accuracy of the delay approximation. We use the mean error (ME) as a function of the steering angle and focal distance (computed at a fixed steering angle θ_i and a fixed focal distance r_j):

$$\mathrm{ME}(i,j) = \sum_{n=1}^{N} [\tau_n^{\mathrm{ex}}(\theta_i, r_j) - \tau_n^{\mathrm{Fr}}(\theta_i, r_i)] / N. \tag{14}$$

The error distribution is important as array elements with larger errors give greater contributions to the beampattern deformation. According to the frequency band of radiated source, an angular region defined by $|\,\theta\,| \le 18^{\circ}$ and a range domain $15 \le r \le 35$ are considered for the computation of ME and the results are shown in Fig. 3.

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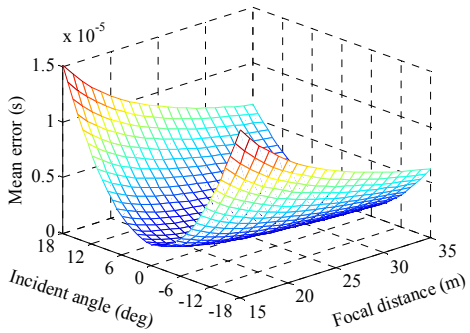
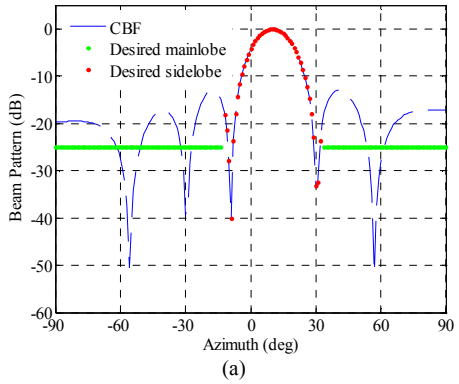


Fig. 3. Mean error surface of the Fresnel delay approximation.

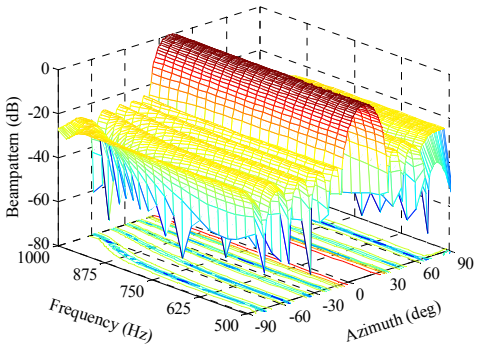
From Fig. 3, it can be seen that the ME originated from the Fresnel approximation increases with incident angle $|\theta|$ and decreases with focal distance r, and its value at the border of the validity region is ME(18°,15m) = 1.5×10^{-5} s. However, within the validity region of the Fresnel approximation, we can affirm that the beampattern deformation produced by the Fresnel approximation is weak and the loss of array gain is generally negligible.

With the Fresnel approximation, the received signal can be compensated with Ψ_n^C and now the design of nearfield FIBs has been transformed into the corresponding farfield implementation situation, and the design method proposed in Sec. III will be used. Here, the design band $[f_L, f_H] = [f_l, f_h]$ is decomposed into M = 22 uniformly distributed subbands. Take the conventional beampattern at reference frequency $f_0 = f_l$ and steering direction of 10° as the desired beampattern. The visible region $\Theta = [-90°, 90°]$ is discretized with a uniform grid of 1°. The mainlobe region is $\Theta_{ML} = [-12°, 32°]$, the sidelobe region is $\Theta_{SL} = [-90°, -14°] \cup [34°, 90°]$ and the prescribed sidelobe level is set to be -25dB, (i.e., $\delta = 0.0562$). The norm constraint is not considered here.

The desired beampattern is shown in Fig. 4(a), and the beampattern variation of the designed farfield FIBs with frequency and azimuth is shown in Fig. 4(b), which demonstrates the perspective frequency invariant characteristics within the mainlobe and the sidelobe level is strictly guaranteed to be below -25 dB.



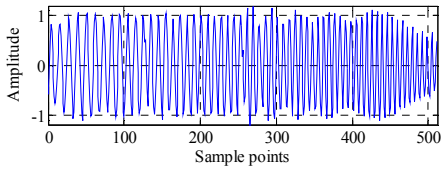
(a)



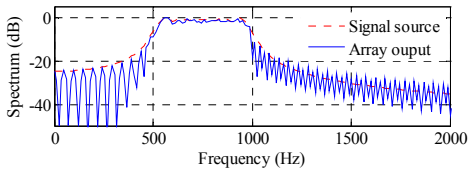
(b)

Fig. 4. Design of the farfield FIBs. (a) The desired beampattern. (b) Variation of beampattern with frequency and azimuth.

Finally, we consider the array output after implementing the aforementioned nearfield focused beamforming. With a 50% overlapping processing of FFT filter-bank, the array output and its corresponding power spectrum are shown in Fig. 5(a) and Fig. 5(b), respectively. From Fig. 5, it can be seen that the results are very encouraging and the time series of array output is very close to the actual waveform of source signal [see Fig. 2(a)]. Notice that there exist two shorter discontinuous variations in array output which is due to the finite data length and finite division of the subbands for beamforming. This can be alleviated by properly improving the subband amount of beamforming with only a little increase in computational loads.



(a)



(b)

Fig. 5. The results of beamforming output. (a) The time series. (b) The power spectrum of source signal and array output.

Although the performance testing is demonstrated for a uniformly spaced linear array, the proposed design scheme can also work for a nonuniform linear array. Future work will be mainly aimed at the application of the proposed design scheme to other array geometries such as planar array with enlarged validity region of the Fresnel approximation.

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